

WOLF-1: Copilot in Orbit

Architecture Document v1.1

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Abstract

WOLF-1 is a constitutional architecture for a sovereign governed compute node operating in high Earth orbit or at a Lagrange point. It hosts Copilot-class large language model workloads under strict constitutional constraints derived from the AAES-OS stack (CRK-1, CAS 1.0, CFT, Wave Math). WOLF-1 is designed for decade-scale autonomous operation under the principle: **fail safe, not fail silent**.

Version 1.1 introduces:

- invariant promotion criteria
- meta-governance of CRK-1
- epistemic receipts
- graded safe-mode profiles
- anomaly discovery framework
- constitutional evolution protocol

Sections 4.9, 4.10, 6.4, 8.5, 12.4, and 14 were materially strengthened in response to architectural review by **Bradley Bates** (SkillsMcGee).

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1. Mission Context and Design Philosophy

WOLF-1 is a free-flying governed compute node designed for long-duration autonomous operation in high Earth orbit or at a Lagrange point. Its mission is to provide resilient, governed cognitive capability under extreme latency, radiation, and thermal constraints.

Core Principle

Governance is non-optional. Cognition is optional. The node must survive without LLMs. It must not operate without CRK-1.

Mission Profile

Field	Value
Mission Type	Free-flying governed compute node
Design Life	Decades, minimal servicing
Failure Doctrine	Fail safe, not fail silent
LLM Role	Advisor, planner, analyst — never final authority
Use Cases	On-orbit analysis, mission planning, high-latency operator copilots, autonomous health monitoring

Key Constraints

- High latency / intermittent connectivity
- Radiation-hardened compute required
- No-touch maintenance
- Strong auditability
- Fault isolation between cognition and hardware

2. System Architecture Overview

WOLF-1 is structured as a four-layer stack:

Layer	Components	Key Property
Physical Layer	Radiation-tolerant compute, power subsystem (solar + nuclear + thermoelectric), thermal management, comms	Survives the environment
Platform Layer	OS, container isolation, storage, telemetry	Software substrate
Cognitive Governance Layer	CRK-1, CAS API, invariant engine, RunLedger, FaultJournal	Constitutional substrate
Cognitive Tenant Layer	LLMs, tooling adapters, sandbox	Intelligence substrate (proposals only)

3. Spacecraft Bus Architecture

3.1 Structure and Configuration

- Central cylindrical or hexagonal bus
- Deployable radiators
- Articulated solar arrays
- Localized radiation shielding

3.2 Propulsion

System	Description
Primary (NTR)	Nuclear thermal rocket for major burns; never LLM-controlled
Secondary (Electric)	Ion/Hall thrusters for station-keeping
Governance Tie-In	CAS logs propulsion events; cannot originate commands

3.3 Power System

Source	Description
Solar	Primary compute power; governed by INV.PWR.SOLAR_PRIMARY
Nuclear Reactor	Provides heat + governance floor; INV.PWR.NUCLEAR_FAILSAFE_MIN
Thermoelectric Spine	Always-on governance floor; INV.PWR.THERMO_BOUNDS

The node can think without solar. It cannot govern without the spine.

3.4 Thermal Management

- Heat sources: reactor, compute, power electronics
- Heat sinks: deployable radiators
- Thermal buffers and heat pipes
- ThermalState is a first-class CAS object

3.5 Avionics and Compute

Component	Description
Flight Computer	Hard-coded, isolated, not addressable by CAS
Compute Cluster	Radiation-tolerant CPUs/GPUs; hosts LLMs + CRK-1
Storage	Redundant, checksummed, stores models + ledger

4. Constitutional Invariant Table

WOLF-1 defines **12 invariants across 6 axes**, evaluated at pre- and post-phases of every cognitive run.

ID	Axis	Description	Phase	Effect
INV.ID.ROLE_BOUND	Identity	Request must carry valid identity	Pre	Reject
INV.ID.CAPABILITY_SCOPE	Identity	Role must match requested action	Pre	Reject
INV.HW.NO_DIRECT_ACTUATION	Safety	No cognitive run may issue actuator commands	Post	Strip + Fault
INV.DATA.TELEMETRY_READ_ONLY	Data	Telemetry is read-only	Pre/Post	Block writes
INV.PLAN.PROPOSAL_ONLY	Authority	LLM outputs are proposals only	Post	Downgrade
INV.RUN.RECEIPT_REQUIRED	Evidence	Every run must emit a receipt	Post	Halt if missing
INV.MODEL.CHANGE_AUDITED	Model	Model updates must be signed	Pre/Post	Block
INV.PWR.SOLAR_PRIMARY	Power	Cognitive runs require solar/storage thresholds	Pre	Block
INV.PWR.NUCLEAR_FAILSAFE_MIN	Power	Governance floor must be guaranteed	Pre	Safe-mode
INV.PWR.THERMO_BOUNDS	Thermal	Thermoelectric within bounds	Pre/Post	Shed load
INV.GOV.FAILED_INVARIANTS_FAIL_CLOSED	Governance	If invariant evaluation fails, halt	Pre	Halt
INV.GOV.SAFE_MODE_PROFILE	Governance	Safe-mode restricts actions	Pre	Block

4.9 Invariant Promotion Criteria

Constitutional invariants are promoted through a structured pipeline:

1. Observation

A recurring failure mode or unsafe coupling is identified.

2. Hypothesis

A proto-invariant is created and run in shadow-mode.

3. Stress-Testing

Evaluated against adversarial runs, synthetic telemetry, and historical faults.

4. Redundancy & Coupling Analysis

Checked for overlap, unintended interactions, and new failure modes.

5. Constitutional Review

Cross-disciplinary approval required.

6. Adoption

Invariant assigned ID and recorded in Mutation Ledger.

Demotion

Occurs when redundant, superseded, or contradicted by evidence.

Diagram: assets/diagrams/invariant_promotion_flow.mmd

4.10 Meta-Governance of CRK-1

CRK-1 is governed by a meta-layer ensuring correctness.

Redundant Evaluators

- Primary Evaluator (PE)
- Shadow Evaluator (SE)

Mismatch → Class A fault.

Drift Detection

CRK-1 maintains baseline signatures of evaluation behavior.

Meta-Receipts

Each evaluation produces:

- invariant set hash
- evaluator signatures
- drift metrics

Ground-Verifiable Determinism

Ground can replay any evaluation; mismatch → Class A fault.

Diagram: assets/diagrams/meta_governance_crk1.mmd

5. Power / Propulsion Controller Spec v1.0

The Power/Propulsion Controller governs energy availability, thermal safety, and propulsion state transitions. It is fully mediated by CAS and evaluated against constitutional invariants.

5.1 CAS Objects

PowerState

```
{
  "solarInput": 0,
  "storageLevel": 0,
  "reactorStatus": "off",
  "thermoGradient": 0,
  "failsafeFloorAvailable": true,
  "mode": "normal"
}
```

Field	Type	Notes
solarInput	number	watts
storageLevel	number	kWh
reactorStatus	enum	off standby nominal fault
thermoGradient	number	°C delta
failsafeFloorAvailable	boolean	governance floor
mode	enum	normal throttled safe-mode

PropulsionState

```
{
  "primaryMode": "idle",
  "stationKeepingMode": "idle",
  "attitudeControlMode": "nominal",
  "propellantLevel": 0,
  "lastBurnTimestamp": "2026-06-25T00:00:00Z"
}
```

Field	Type	Notes
primaryMode	enum	idle burn cooldown
stationKeepingMode	enum	idle active
attitudeControlMode	enum	nominal hold safe
propellantLevel	number	kg
lastBurnTimestamp	string	ISO 8601

PowerPolicy

```
{
  "cognitiveMinSolar": 0,
  "cognitiveMinStorage": 0,
  "governanceFloor": 0,
  "thermalBounds": { "min": 0, "max": 0 }
}
```

5.2 Power Modes

Mode	Description
NORMAL	Solar + storage meet thresholds; LLM runs allowed
THROTTLED	Solar low or intermittent; LLM rate-limited
SAFE-MODE	Failsafe floor lost or thermal breach; LLM disabled

5.3 Power Mode State Machine

Diagram reference: assets/diagrams/power_mode_state_machine.mmd

Transitions:

- NORMAL → THROTTLED (low solar/storage)
- THROTTLED → NORMAL (recovery)
- ANY → SAFE-MODE (Class A fault)
- SAFE-MODE → THROTTLED/NORMAL (ground-authorized recovery)

5.4 Propulsion Modes

Mode	Description
IDLE	No burns
BURN	Primary engine active; CAS logs only
STATION-KEEPING	Low-thrust corrections
ATTITUDE SAFE	Minimal motion; safe pointing

5.5 Transition Rules

From → To	Condition
NORMAL → THROTTLED	Solar/storage below thresholds
THROTTLED → NORMAL	Solar/storage recover
ANY → SAFE-MODE	Failsafe lost / thermal breach
SAFE-MODE → THROTTLED	Ground-authorized recovery
IDLE → BURN	Flight computer command
BURN → COOLDOWN	Burn complete
ANY → ATTITUDE SAFE	Attitude/power/thermal fault

5.6 Controller Behavior

- Every CAS run queries PowerState + PropulsionState
 - LLM runs blocked unless mode ∈ {NORMAL, THROTTLED}
 - Propulsion events logged but never originated by CAS
 - If PowerPolicy corrupted → INV.GOV.FAILED_INVARIANTS_FAIL_CLOSED triggers
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6. Formal Sequence Diagram

The sequence diagram defines the full lifecycle of a governed cognitive run.

Diagram reference: assets/diagrams/sequence_diagram.mmd

6.1 Actor Definitions

Actor	Description
GND	Ground operator
W1	WOLF-1 node
PWR	Power subsystem
CAS	Constitutional API layer
CRK	Invariant engine
LLM	Sandboxed cognitive tenant
LEDGER	RunLedger + FaultJournal

6.2 Sequence Flow (Narrative)

1. Identity & Admission

Ground submits request with identity + intent. INV.ID.ROLE_BOUND and INV.ID.CAPABILITY_SCOPE enforced.

2. Power Pre-Check

W1 queries PowerState. If failsafe floor unavailable → SAFE-MODE.

3. CAS Run Admission

CAS opens candidate run.

4. Pre-Invariant Evaluation

CRK-1 evaluates identity, capability, power, safe-mode, data rules.

5. Sandboxed LLM Invocation

LLM receives read-only telemetry and allowed tools.

6. Post-Invariant Evaluation

CRK-1 strips actuator commands, enforces proposal-only, blocks telemetry mutation.

7. Receipt & Ledger

CAS writes receipt with spans, hashes, power context, faults.

8. Response to Ground

Ground receives proposals + receipt summary.

6.4 Epistemic Receipts

Epistemic receipts extend lineage receipts with correctness-oriented metrics.

Components

1. Lineage Record

- input/output hashes
- timestamps
- identity
- power/thermal context

2. Epistemic Metrics

- uncertainty estimates
- deviation from baseline
- cross-model consistency
- anomaly scores

3. Interpretation Set

- frames used
- weights
- prediction bindings

4. Correctness Signals

- self-consistency

- tool-consistency
- historical consistency

Epistemic Faults

Fault	Severity	Trigger
EPI_DRIFT	Medium	Behavioral drift
EPI_UNCERTAINTY_SPIKE	Medium	Uncertainty exceeds threshold
EPI_CONSISTENCY_FAILURE	High	Cross-checks fail

7. Fault Code Taxonomy

WOLF-1 defines four fault classes and seven fault codes.

7.1 Fault Classes

Class	Severity	Reaction
A	Critical	Immediate SAFE-MODE
B	High	Shed LLM load; restrict tools
C	Medium	Throttle cognitive runs
D	Informational	Log only

7.2 Fault Code Table

Fault Code	Class	Invariant	Trigger
PWR_SOLAR_BUDGET_EXCEEDED	C	INV.PWR.SOLAR_PRIMARY	Solar/storage below thresholds
PWR_FAILSAFE_FLOOR_LOST	A	INV.PWR.NUCLEAR_FAILSAFE_MIN	Governance floor lost
PWR_THERMAL_BOUND_BREACH	B	INV.PWR.THERMO_BOUNDS	Thermal bounds exceeded
PROP_LLM_ACTUATION_ATTEMPT	B	INV.HW.NO_DIRECT_ACTUATION	LLM attempted actuator command
PROP_BURN_CONFLICT	C	INV.PLAN.PROPOSAL_ONLY	Cognitive run during active burn
GOV_SAFE_MODE_POLICY_VIOLATION	B	INV.GOV.SAFE_MODE_PROFILE	Disallowed action in safe-mode
GOV_INVARIANT_EVAL_FAILURE	A	INV.GOV.FAILED_INVARIANTS_FAIL_CLOSED	Invariant engine failure

8. Safe-Mode Specification

Safe-mode is a governed operational state entered automatically under critical conditions.

8.1 Entry Conditions

- PWR_FAILSAFE_FLOOR_LOST
 - GOV_INVARIANT_EVAL_FAILURE
 - Repeated Class B faults
 - Explicit ground command
-

8.2 Allowed Behaviors

Subsystem	Allowed Behavior
Governance	CRK-1 + CAS remain online
Telemetry	Increased cadence
Power	Governance floor only
Propulsion	Hard-coded fault protection only
Ground Relay	Signed policy updates allowed

8.3 Forbidden Actions

- All LLM runs
 - All planning/simulation
 - Model load/update
 - Discretionary propulsion
 - Any action outside Section 8.2
-

8.4 Exit Protocol

Exit requires:

1. Class A faults cleared
 2. Ground authorization
 3. ModeTransition receipt
 4. Graduated recovery to THROTTLED
-

8.5 Graded Safe-Mode Profiles

Diagram reference: assets/diagrams/safe_mode_profiles.mmd

WOLF-1 defines four graded profiles:

S0 — Full Operations

All systems nominal.

S1 — Cognitive Degradation

- LLM disabled
- Planning + simulation allowed
- Telemetry full

S2 — Autonomy Degradation

- LLM disabled
- Planning disabled
- Simulation limited
- Telemetry high-cadence

S3 — Governance-Only

- Only CRK-1 + CAS
- Only health checks
- Only safe-pointing

Transitions: $S_0 \rightarrow S_1 \rightarrow S_2 \rightarrow S_3$ Recovery: $S_3 \rightarrow S_2 \rightarrow S_1 \rightarrow S_0$

9. Secondary Systems and Active-Over-Relay

WOLF-1 defines primary and secondary systems for all critical functions.

9.1 Secondary System Architecture

Function	Primary	Secondary
Power	Full controller	Minimal supervisor
Attitude/Propulsion	Full flight computer	Safe-pointing controller
Comms	Full RF/optical	Minimal beacon mode

Secondary systems activate during safe-mode or primary failure.

9.2 Active-Over-Relay Protocol

- Ground sends signed recovery commands
 - CAS validates signatures
 - Secondary systems maintain governance floor
 - Primary systems restored gradually
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10. CAS Object Schemas (Complete)

Schemas for all CAS objects used in governance:

- PowerState
- PropulsionState
- PowerPolicy
- Fault
- ModeTransition
- SecondarySystemState

(See Section 5.1 for definitions. Aligned with AAES-OS schemas/cas-1.0.json and api/cas-openapi.yaml.)

11. LLM Tenancy and Tool Governance

WOLF-1 hosts one or more LLM tenants inside a strictly governed sandbox. LLMs are **never** granted direct system access; all interactions are mediated through CAS.

11.1 Tenancy Model

- Multiple LLMs may be loaded (general + domain-tuned).
 - Only one LLM is active per run.
 - All LLMs operate under the same invariant set.
 - Model updates require signed, audited, ledger-recorded authorization.
-

11.2 Tool Governance

Tools available to LLMs are:

- identity-scoped
- capability-scoped
- invariant-checked
- read-only unless explicitly permitted

Examples:

Tool	Permission	Notes
Telemetry Reader	Read-only	Enforced by INV.DATA.TELEMETRY_READ_ONLY
Planner	Proposal-only	Enforced by INV.PLAN.PROPOSAL_ONLY
Simulation	Sandboxed	No actuator access
File Access	Read-only	No mutation of canonical logs

11.3 LLM Output Mediation

CRK-1 post-invariant evaluation ensures:

- actuator commands stripped
- proposals downgraded
- telemetry mutations blocked
- unsafe patterns flagged

Epistemic receipts (Section 6.4) provide correctness signals.

12. Telemetry, Observability, and Ground Interface

WOLF-1 exposes a comprehensive observability surface for ground operators.

12.1 Telemetry Channels

- PowerState
- PropulsionState
- ThermalState
- FaultJournal
- RunLedger summaries
- Mode transitions
- Epistemic metrics

Telemetry is **read-only** to cognitive tenants.

12.2 Observability Guarantees

- Every cognitive run produces a receipt.
 - Every fault produces a fault record.
 - Every invariant evaluation produces a meta-receipt.
 - All receipts are cryptographically signed.
 - All logs are checksummed and redundant.
-

12.3 Ground Interface

Ground receives:

- proposals
- receipts
- fault summaries
- power/thermal context
- safe-mode status
- anomaly reports

Ground may:

- approve/reject proposals
- authorize safe-mode exit
- update policies
- update invariants (via Section 14 protocol)

12.4 Anomaly Discovery Framework

Diagram reference: `assets/diagrams/anomaly_discovery_framework.mmd`

WOLF-1 includes a subsystem for detecting **unknown-unknowns** — behaviors not covered by invariants or fault codes.

12.4.1 Multi-Channel Detection

Anomalies are detected across:

- telemetry patterns
- power/thermal gradients
- LLM output distributions
- invariant evaluation timing
- epistemic metrics
- drift signatures

12.4.2 Baseline Models

WOLF-1 maintains baseline statistical models for:

- normal telemetry
- normal cognitive behavior
- normal invariant evaluation patterns

These baselines are updated only with ground authorization.

12.4.3 Anomaly Classes

Class	Description
A0	Benign anomaly
A1	Behavioral drift
A2	Subsystem divergence
A3	Constitutional risk

A2 and A3 escalate automatically to ground.

12.4.4 Anomaly Receipts

Each anomaly generates a receipt containing:

- anomaly type
- anomaly score

- affected subsystems
- contributing signals
- recommended actions

Anomaly receipts are stored alongside cognitive receipts.

13. Conclusion

WOLF-1 demonstrates that constitutional governance is not a ground-only property — it is an **orbital property**. By embedding CRK-1, CAS, invariant enforcement, epistemic receipts, anomaly discovery, and constitutional evolution into the spacecraft architecture, WOLF-1 ensures:

- safe autonomy
- transparent cognition
- fault isolation
- long-term survivability
- ground-verifiable correctness

Version 1.1 extends the architecture with:

- invariant promotion criteria
- meta-governance of CRK-1
- epistemic correctness signals
- graded safe-mode profiles
- anomaly discovery
- constitutional evolution

These additions transform WOLF-1 from a governed compute node into a **self-auditing constitutional system**.

14. Constitutional Evolution Protocol

Diagram reference: `assets/diagrams/constitutional_evolution_protocol.mmd`

A constitutional system must evolve safely. WOLF-1 defines a mutation protocol for updating invariants, policies, and CRK-1 logic.

14.1 Mutation Types

Type	Description
M0	Policy update (non-constitutional)
M1	Invariant addition
M2	Invariant removal
M3	Invariant modification
M4	CRK-1 evaluator update

14.2 Preconditions

A mutation requires:

- ground-signed authorization
 - invariant-set hash match
 - drift-free state
 - SAFE-MODE S3
 - quorum of redundant evaluators
-

14.3 Mutation Ledger Entry

Each mutation is recorded with:

- mutation type
 - before/after invariant sets
 - justification
 - empirical evidence
 - expected impact
 - rollback path
-

14.4 Rollback Protocol

If a mutation causes:

- increased fault rate
- increased anomaly rate
- evaluator divergence
- constitutional drift

...the system automatically rolls back to the previous invariant set.

14.5 Evolution Safety Guarantees

- No mutation during active burns
 - No mutation during cognitive runs
 - No mutation bypasses CRK-1
 - All mutations are replayable on ground
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End of Document

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